

Koppuravuri Himavanth Ayyappa

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B.Tech CSD student with a strong foundation in AI/ML, Generative AI, and full-stack software development. Experienced in building RESTful APIs, working with **LLMs**, **RAG pipelines**, and **agentic AI** tools, alongside hands-on exposure to high-performance/**GPU computing** and **performance optimization**. Passionate about applying AI/ML to solve real-world engineering problems through collaborative, **scalable software development**.

Technical Skills

Programming Languages: Python, Java, C, C++
AI / Machine Learning: LLMs, LLMops, MLOps, RAG, LangChain, Prompt Engineering, Fine tuning
Agentic AI Tools: Cursor, Claude Code, GitHub Copilot
Core CS: Data Structures & Algorithms, OOPS, Operating Systems, DBMS
Frontend Technologies: HTML, CSS, React.js, vue.js
Backend Technologies: Node.js, Express.js, FastAPI, Flask, REST APIs
Databases: MongoDB, SQL, Vector Databases
Tools & Platforms: Linux, Git, GitHub, Jenkins, Docker, AWS

Education

Keshav Memorial Institute of Technology, Hyderabad	2023–2027
B.Tech – CSD (Computer Science and Data Science)	CGPA: 8.55
IIT Madras	2023–2027
Online Diploma in Programming	CGPA: 8
Narayana Junior College	2021–2023
Board of Intermediate Education, Telangana – MPC	Percentage: 93
Narayana Concept School	2021
Board of Secondary Education, Telangana – SSC	CGPA: 10

Experience

Defence Research Development Laboratory – DRDL	Mar 2025 – Feb 2026
HPC Intern	
- Accelerated CPU workloads using CUDA C, achieving up to 91.7% runtime reduction (12x) through efficient GPU parallelization. - Implemented and evaluated NVIDIA MPS to enable concurrent multi-process GPU execution, improving SM (GPU core) utilization and throughput. - Conducted 50+ performance profiling, debugging, and kernel optimization sessions using tools like gprof, ensuring 100% execution stability for Reynold's Equations. - Performed HPC scaling and performance analysis across multi-node, multi-GPU configurations, identifying node topology and MPI communication overhead as key factors affecting scalability, and used Python to automate benchmarking and performance visualization.	

Projects

AI Document Assistant – RAG-based Q&A System	Mar 2026 - Apr 2026
- Built a full-stack Generative AI application with a FastAPI backend and React frontend, enabling natural-language Q&A over user-uploaded documents. - Implemented a Retrieval-Augmented Generation (RAG) pipeline using LangChain and a vector database (ChromaDB/FAISS) for semantic search and context retrieval. - Integrated open-source LLMs via Hugging Face for context-aware response generation, and designed agentic workflows (LangChain agents) for multi-step task automation including document parsing, retrieval, and summarization. Tools Used: Python, FastAPI, LangChain, ChromaDB, React, Hugging Face Transformers	
Pocket LLM	Feb 2025 - May 2025
Built a fully offline mobile AI assistant, supporting 5+ core features including calls, SMS, alarms, and reminders.	

- Enabled 100% offline LLM inference using **llama.cpp via JNI**, achieving sub-2s response time on-device with quantized GGUF models, improving privacy compliance.
- Tested across 10+ real user queries, maintaining consistent conversational accuracy.
- **Tools Used:** Kotlin, JNI, llama.cpp, Orca-Mini-3B (LLM)

Dyslexic Kids | GitHub

Sep 2024 - Dec 2024

- Built a real-time facial expression tracking system for dyslexic children, integrating a **ViT model from Hugging Face** for facial expression analysis with **92% accuracy**, personalizing therapy sessions.
- Designed backend services using a **microservices-inspired architecture**, with a low-latency **Node.js/Express** pipeline to process video frames and stream emotion updates via REST APIs.
- Designed secure **authentication** and **authorization** systems with **idempotency** and **rate limiting**, ensuring reliable operations on therapist dashboards.
- **Tools Used:** MERN stack, ViT Face Expression Tracker (Hugging Face)

Leadership & Activities

Member, Recurse – Technical Club of KMIT

June 2024 – Present

- Active member of Recurse, the technical club of KMIT, contributing to technical and innovative initiatives.
- Assisted in planning and executing technical events, hackathons, and coding competitions.
- Actively participated in team-building and knowledge-sharing initiatives.

Achievements

- **Finalist – Smart India Hackathon 2024** (National level)
- **Finalist – Green Tech Hackathon 2025** at Vasavi College among 80+ teams